

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: VOLUME AND CAPACITY (SUB-STRAND 2.7)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation of the Kericho containers puzzle. Mr. Kiprop asked Mama Chebet why his bucket is not as big as her drum when they look the same size. You will now answer him using mathematics. Use $\pi = 22/7$ in all calculations. Remember: $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$ and $1 \text{ m}^3 = 1000 \text{ L}$. Write your answers in the space under each prompt. Show every step of your working, include units, and give litres to 2 decimal places for the Kericho containers. Refer to the containers as drum (A), cover (B) and bucket (C). Use the evidence you collected in Lessons 1 to 8 and link every answer back to the driving question.

Part 1: Same Look, Different Shape

Mama Chebet's drum (A), conical sisal cover (B) and metal bucket (C) are all about 60 cm tall with a 42 cm opening. (a) Write down the opening radius. (b) Explain why having the same height and the same opening does not mean the containers hold the same amount. (c) Think back to your Lesson 1 prediction. How did your predicted order compare with what you later found, and what idea were you missing at that time? (d) State the difference between volume and capacity, with the units used for each.

- (a) The opening diameter is 42 cm, so the radius is $42 \div 2 = 21 \text{ cm}$.
- (b) Height and opening only describe the size of the top and how tall the container is. They do not describe how the sides behave between the top and the bottom. The drum keeps the same width all the way down, the cover narrows steadily to a point, and the bucket narrows only to a smaller bottom. Because the shapes are different, the space inside is different, so shape decides volume.
- (c) Answers will vary. A good answer states the order that was predicted (for example, many learners expected the three to be nearly equal or the bucket to be the biggest) and says that the missing idea was how the shape (tapering) changes volume, for which we needed the formulas for the cone and the frustum. The actual order is cover B < bucket C < drum A.
- (d) Volume is the space inside a solid and is measured in cm^3 or m^3 . Capacity is the amount a container can hold and is measured in mL or L. They are linked by $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$ and $1 \text{ m}^3 = 1000 \text{ L}$.

Part 2: The Drum and the Cover

(a) Describe the pouring investigation you did with a hollow cone and a hollow cylinder of the same base

- (a) We filled the cone to the brim and poured it into the cylinder with the same base radius and height. The cylinder was full after exactly 3 cone-fuls, in every trial and every group.
- (b) $V(\text{cone}) = \frac{1}{3} \pi r^2 h$, which is one third of the cylinder $V = \pi r^2 h$.

radius and height. What did you find? (b) Write the formula for the volume of a cone. (c) Calculate the volume of the cylindrical drum (A) and of the conical cover (B) in cm ³ and then in litres. (d) Show how your answers prove the relationship between the drum and the cover.	(c) Drum A: $V = (22/7) \times 21 \times 21 \times 60 = 83,160 \text{ cm}^3 = 83.16 \text{ L}$. Cover B: $V = 1/3 \times 83,160 = 27,720 \text{ cm}^3 = 27.72 \text{ L}$. (d) $83.16 \div 27.72 = 3$, so the drum holds exactly 3 times as much as the cover. This matches the 3 cone-fuls that filled the cylinder. The cover tapers to a point, so it holds only one third of the drum.
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Part 3: Explaining the Bucket

Bucket C has top radius R = 21 cm, bottom radius r = 14 cm and height h = 60 cm. (a) Explain how a frustum of a cone can be thought of in terms of two cones. (b) Use similar triangles to find the height of the small cone that was removed and of the full cone. (c) Calculate the volume of the full cone and of the small cone. (d) Find the capacity of the bucket in litres and check it with the frustum formula. (e) Answer Mr. Kiprop: why is his bucket not as big as the drum, yet bigger than the cover? Use your numbers.	(a) The bucket is a large cone (height 180 cm) with the small tip cone (height 120 cm) removed from the pointed end, so $V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})$. Extending the slanting sides until they meet at a point below the narrow bottom of the bucket shows the full cone. (b) The ratio $r : R = 14 : 21 = 2 : 3$. Let the small cone have height x. Then $x/(x + 60) = 2/3$, so $3x = 2x + 120$ and $x = 120 \text{ cm}$. The full cone has height $120 + 60 = 180 \text{ cm}$. (c) Large cone: $1/3 \times (22/7) \times 21^2 \times 180 = 83,160 \text{ cm}^3$. Small cone: $1/3 \times (22/7) \times 14^2 \times 120 = 24,640 \text{ cm}^3$. (d) Bucket = $83,160 - 24,640 = 58,520 \text{ cm}^3 = 58.52 \text{ L}$. Check: $V = 1/3 \pi h (R^2 + Rr + r^2) = 1/3 \times (22/7) \times 60 \times (441 + 294 + 196) = 1/3 \times (22/7) \times 60 \times 931 = 58,520 \text{ cm}^3$. The two methods agree. (e) Mr. Kiprop, your bucket holds 58.52 L. The drum holds 83.16 L because it does not taper at all. The cover holds 27.72 L because it tapers all the way to a point. The bucket tapers only to a 28 cm base, so it holds more than the cover and less than the drum (about 70% of the drum). The order is cover 27.72 L < bucket 58.52 L < drum 83.16 L.
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Part 4: Pyramids and Their Frustums

(a) A hollow square-based pyramid with base 30 cm × 30 cm and height 40 cm was poured into a cuboid of the same base and height. State what you found and write the pyramid volume rule. Calculate the pyramid's volume in cm³ and litres. (b) A maize bin for Rift Valley farmers is a square-	(a) The pyramid filled the cuboid exactly 3 times, so the pyramid is one third of the cuboid. $V = 1/3 \times \text{base area} \times \text{height} = 1/3 \times 900 \times 40 = 12,000 \text{ cm}^3 = 12 \text{ L}$ (the cuboid is 36,000 cm³). (b) $A_1 = 60 \times 60 = 3,600 \text{ cm}^2$, $A_2 = 30 \times 30 = 900 \text{ cm}^2$, $\text{sqrt}(A_1 A_2) = \text{sqrt}(3,240,000) = 1,800 \text{ cm}^2$. $V = 1/3 \times h \times (A_1 + A_2 + \text{sqrt}(A_1 A_2)) = 1/3 \times 40 \times (3,600 + 900 + 1,800) = 1/3 \times 40 \times 6,300 = 84,000 \text{ cm}^3 = 84 \text{ L}$. (c) The large pyramid has side 60 cm and height 80 cm: $1/3 \times 3,600 \times 80 = 96,000 \text{ cm}^3$. The small pyramid has side 30 cm and height 40 cm: $12,000 \text{ cm}^3$. $96,000 - 12,000 = 84,000 \text{ cm}^3 = 84 \text{ L}$, which matches (b). (d) Mass = $84 \times 0.75 = 63 \text{ kg}$ of maize.
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based frustum with top side 60 cm, bottom side 30 cm and height 40 cm. Calculate its capacity in litres using the frustum formula for a pyramid. (c) Check your answer using the large-minus-small pyramid method. (d) The bulk density of maize is 0.75 kg per litre. Find the mass of maize the bin holds.	
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Part 5: Designing and Using Mathematics	
(a) A cup is a frustum with top radius 6 cm, bottom radius 3 cm and height 7 cm. Calculate its volume, then find how many such cups fill the Kericho drum (A). (b) When you built and tested your own model containers, your measured capacity was not exactly the same as the calculated one. Explain how you calculated percentage error and give two reasons for the difference. (c) Write a short final explanation (4 to 6 sentences) that answers the driving question: why do the three containers hold such different amounts, and how can we use mathematics to design a frustum container that holds exactly the capacity we need? Include the key numbers. (d) Using your answer to the KICD key inquiry question, explain why knowledge of volume and capacity is useful.	(a) $V = \frac{1}{3} \times (22/7) \times 7 \times (36 + 18 + 9) = \frac{1}{3} \times 22 \times 63 = 462 \text{ cm}^3$. Cups per drum = $83,160 \div 462 = 180$ cups. (b) Percentage error = (measured - calculated) $\div$ calculated $\times 100\%$, taken as a size without regard to sign. Our errors were about 5-10%. Reasons: the thickness of the card, gaps or imperfect seams, models that are not perfectly shaped, and water or sand that was spilled when measuring. (c) The drum (A), cover (B) and bucket (C) look alike because they are all about 60 cm tall with a 42 cm opening (radius 21 cm), but their shapes are different and shape decides volume. The cone holds exactly one third of the cylinder, $V = \frac{1}{3} \pi r^2 h$, so cover B holds 27.72 L against 83.16 L for drum A. The bucket is a large cone ($83,160 \text{ cm}^3$) with a small cone ($24,640 \text{ cm}^3$) removed, so it holds 58.52 L, between the two. For a frustum, $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$, and for a pyramid frustum $V = \frac{1}{3} h (A1 + A2 + \text{sqrt}(A1 A2))$. To design a frustum container, we choose the required capacity, pick the dimensions, calculate the volume with the formula, convert to litres and adjust the dimensions until the answer matches. (d) Volume and capacity knowledge is useful because it lets us design and choose containers that hold the amount we need, such as tea storage in Kericho, a 6,160 L conical water tank for a Turkana homestead, or a githeri pot of 80.52 L that can serve 100 people at 0.5 L each. It also lets us avoid waste, check that something will fit, and convert capacity into mass, such as 63 kg of maize in the bin.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)

Mathematical accuracy and units	All volumes, capacities, ratios and conversions are correct, use $\pi = 22/7$, and include correct units with litres to 2 decimal places for the Kericho containers.	Most calculations are correct, with one or two small slips in arithmetic or units that do not change the conclusion.	Several calculations or conversions are incorrect, or units are missing, so the conclusions are not supported.
Use of formulas and methods	Chooses the correct formula for each solid and uses the large-minus-small method and similar triangles accurately, showing the cone-cylinder and pyramid-cuboid one-third relationships.	Uses the correct formulas in most cases but the working for similar triangles or large-minus-small is partly incomplete.	Uses wrong formulas or cannot show how a frustum relates to a cone or pyramid.
Explaining the phenomenon	Clearly explains that the containers differ in capacity because their shape differs, uses 27.72 L, 58.52 L and 83.16 L, and answers Mr. Kiprop's question with the tapering reasoning.	Explains that shape matters and quotes the capacities, but the reasoning about tapering is general or incomplete.	States that the containers differ without a mathematical explanation or with incorrect numbers.
Evidence, error and design reasoning	Uses evidence from the pouring and model investigations, explains percentage error and its causes, and describes a clear process for designing a frustum of a required capacity.	Mentions evidence and at least one cause of error, and gives a partial design process.	Gives little evidence from the investigations and no clear explanation of error or design.
Communication and usefulness of volume and capacity	Writes clear, well-organised steps in sensible language and gives well-chosen Kenyan examples that show why volume and capacity are useful.	Writing is mostly clear and gives one or two examples of usefulness.	Writing is hard to follow and gives no examples of why volume and capacity are useful.

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